Title Page

This section provides a comprehensive analytical explanation of the step being performed in the data pipeline. The process begins with understanding the structure and semantics of the dataset, ensuring that all variables are correctly interpreted before any transformation is applied. Each operation is carefully designed to improve data quality, enhance analytical capability, and align the dataset with business objectives. For instance, data cleaning steps such as handling missing values and standardizing column names ensure consistency and prevent errors during downstream SQL queries. Feature engineering techniques, such as creating age groups, enable more meaningful segmentation and support targeted business strategies. From a technical perspective, these transformations improve computational efficiency and allow for optimized query execution. From a business standpoint, they enable deeper insights into customer behavior, purchasing patterns, and revenue drivers. The integration of Python for preprocessing, SQL for structured analysis, and Power BI for visualization represents a robust end-to-end analytics pipeline. Each stage contributes to converting raw transactional data into actionable intelligence, facilitating informed decision-making and strategic planning in a real-world business environment.

Certificate / Declaration

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Acknowledgement

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Abstract

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Introduction

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Problem Statement & Business Context

This section provides a comprehensive analytical explanation of the step being performed in the data pipeline. The process begins with understanding the structure and semantics of the dataset, ensuring that all variables are correctly interpreted before any transformation is applied. Each operation is carefully designed to improve data quality, enhance analytical capability, and align the dataset with business objectives. For instance, data cleaning steps such as handling missing values and standardizing column names ensure consistency and prevent errors during downstream SQL queries. Feature engineering techniques, such as creating age groups, enable more meaningful segmentation and support targeted business strategies. From a technical perspective, these transformations improve computational efficiency and allow for optimized query execution. From a business standpoint, they enable deeper insights into customer behavior, purchasing patterns, and revenue drivers. The integration of Python for preprocessing, SQL for structured analysis, and Power BI for visualization represents a robust end-to-end analytics pipeline. Each stage contributes to converting raw transactional data into actionable intelligence, facilitating informed decision-making and strategic planning in a real-world business environment.

Objectives

This section provides a comprehensive analytical explanation of the step being performed in the data pipeline. The process begins with understanding the structure and semantics of the dataset, ensuring that all variables are correctly interpreted before any transformation is applied. Each operation is carefully designed to improve data quality, enhance analytical capability, and align the dataset with business objectives. For instance, data cleaning steps such as handling missing values and standardizing column names ensure consistency and prevent errors during downstream SQL queries. Feature engineering techniques, such as creating age groups, enable more meaningful segmentation and support targeted business strategies. From a technical perspective, these transformations improve computational efficiency and allow for optimized query execution. From a business standpoint, they enable deeper insights into customer behavior, purchasing patterns, and revenue drivers. The integration of Python for preprocessing, SQL for structured analysis, and Power BI for visualization represents a robust end-to-end analytics pipeline. Each stage contributes to converting raw transactional data into actionable intelligence, facilitating informed decision-making and strategic planning in a real-world business environment.

Tools & Technologies Used

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Dataset Description

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Python Data Loading & Preview

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[4]:
```
import pandas as pd
df = pd.read_csv('customer_shopping_behaviour.csv')
```

[5]:
```
df.head()
```

[5]:

Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo	Fortnightly
Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash	Fortnightly
Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card	Weekly
Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal	Weekly
Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal	Annually

Figure: Relevant screenshot with explanation of transformation and output.

Data Structure Inspection

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[7]:
```
df.describe()
```

[7]:

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

Figure: Relevant screenshot with explanation of transformation and output.

Statistical Summary

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```
[8]: df.isnull().sum()
```

```
[8]: Customer ID                0
     Age                        0
     Gender                     0
     Item Purchased             0
     Category                   0
     Purchase Amount (USD)      0
     Location                   0
     Size                       0
     Color                      0
     Season                     0
     Review Rating             37
     Subscription Status        0
     Shipping Type              0
     Discount Applied           0
     Promo Code Used            0
     Previous Purchases         0
     Payment Method             0
     Frequency of Purchases     0
     dtype: int64
```

```
[10]: df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
```

Figure: Relevant screenshot with explanation of transformation and output.

Missing Value Detection

This section provides a comprehensive analytical explanation of the step being performed in the data pipeline. The process begins with understanding the structure and semantics of the dataset, ensuring that all variables are correctly interpreted before any transformation is applied. Each operation is carefully designed to improve data quality, enhance analytical capability, and align the dataset with business objectives. For instance, data cleaning steps such as handling missing values and standardizing column names ensure consistency and prevent errors during downstream SQL queries. Feature engineering techniques, such as creating age groups, enable more meaningful segmentation and support targeted business strategies. From a technical perspective, these transformations improve computational efficiency and allow for optimized query execution. From a business standpoint, they enable deeper insights into customer behavior, purchasing patterns, and revenue drivers. The integration of Python for preprocessing, SQL for structured analysis, and Power BI for visualization represents a robust end-to-end analytics pipeline. Each stage contributes to converting raw transactional data into actionable intelligence, facilitating informed decision-making and strategic planning in a real-world business environment.

[11]:
```
df.isnull().sum()
```

[11]:
```
Customer ID               0
Age                       0
Gender                    0
Item Purchased            0
Category                  0
Purchase Amount (USD)     0
Location                  0
Size                      0
Color                     0
Season                    0
Review Rating             0
Subscription Status       0
Shipping Type             0
Discount Applied          0
Promo Code Used           0
Previous Purchases        0
Payment Method            0
Frequency of Purchases    0
dtype: int64
```

[34]:
```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ','_')
df = df.rename(columns = {'purchase_amount_(usd)':'purchase_amount'})
```

Figure: Relevant screenshot with explanation of transformation and output.

Missing Value Handling

This section provides a comprehensive analytical explanation of the step being performed in the data pipeline. The process begins with understanding the structure and semantics of the dataset, ensuring that all variables are correctly interpreted before any transformation is applied. Each operation is carefully designed to improve data quality, enhance analytical capability, and align the dataset with business objectives. For instance, data cleaning steps such as handling missing values and standardizing column names ensure consistency and prevent errors during downstream SQL queries. Feature engineering techniques, such as creating age groups, enable more meaningful segmentation and support targeted business strategies. From a technical perspective, these transformations improve computational efficiency and allow for optimized query execution. From a business standpoint, they enable deeper insights into customer behavior, purchasing patterns, and revenue drivers. The integration of Python for preprocessing, SQL for structured analysis, and Power BI for visualization represents a robust end-to-end analytics pipeline. Each stage contributes to converting raw transactional data into actionable intelligence, facilitating informed decision-making and strategic planning in a real-world business environment.

[35]:
```
df.columns
```

[35]:
```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'promo_code_used', 'previous_purchases',
       'payment_method', 'frequency_of_purchases'],
      dtype='object')
```

[37]:
```
# create a column age_group
labels = ['young_adult' , 'adult' , 'middle_age' , 'senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels =labels)
```

[38]:
```
df[['age','age_group']].head()
```

[38]:

	age	age_group
0	55	middle_age
1	19	young_adult
2	50	middle_age
3	21	young_adult
4	45	middle_age

Figure: Relevant screenshot with explanation of transformation and output.

Column Standardization

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[52]:
```
df[['discount_applied','promo_code_used']].head(6)
```

[52]:

	discount_applied	promo_code_used
0	Yes	Yes
1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes
5	Yes	Yes

[54]:
```
(df['discount_applied'] == df['promo_code_used']).all()
```

[54]:
```
np.True_
```

[55]:
```
df = df.drop('promo_code_used',axis=1)
```

Figure: Relevant screenshot with explanation of transformation and output.

Feature Engineering

This section provides a comprehensive analytical explanation of the step being performed in the data pipeline. The process begins with understanding the structure and semantics of the dataset, ensuring that all variables are correctly interpreted before any transformation is applied. Each operation is carefully designed to improve data quality, enhance analytical capability, and align the dataset with business objectives. For instance, data cleaning steps such as handling missing values and standardizing column names ensure consistency and prevent errors during downstream SQL queries. Feature engineering techniques, such as creating age groups, enable more meaningful segmentation and support targeted business strategies. From a technical perspective, these transformations improve computational efficiency and allow for optimized query execution. From a business standpoint, they enable deeper insights into customer behavior, purchasing patterns, and revenue drivers. The integration of Python for preprocessing, SQL for structured analysis, and Power BI for visualization represents a robust end-to-end analytics pipeline. Each stage contributes to converting raw transactional data into actionable intelligence, facilitating informed decision-making and strategic planning in a real-world business environment.

```
[55]: df = df.drop('promo_code_used',axis=1)

[56]: df.columns

[56]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'previous_purchases', 'payment_method',
       'frequency_of_purchases', 'age_group', 'purchase_frequency_days',
       'purchase_frequency'],
      dtype='object')
```

Figure: Relevant screenshot with explanation of transformation and output.

Redundant Column Removal

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[58]:
```
pip install psycopg2-binary sqlalchemy
```

```
Requirement already satisfied: psycopg2-binary in c:\users\admin\anaconda3\lib\site-packages (2.9.12)
Requirement already satisfied: sqlalchemy in c:\users\admin\anaconda3\lib\site-packages (2.0.43)
Requirement already satisfied: greenlet>=1 in c:\users\admin\anaconda3\lib\site-packages (from sqlalchemy)
Requirement already satisfied: typing-extensions>=4.6.0 in c:\users\admin\anaconda3\lib\site-packages (frc
Note: you may need to restart the kernel to use updated packages.
```

[64]:
```
from sqlalchemy import create_engine

username = "postgres"
password = "2892004"
host = "localhost"
port = "5432"
database = "Customer_behaviour"

engine = create_engine(f"postgresql+psycopg2://{username}:{password}@{host}:{port}/{database}")

table_name = "customer"
df.to_sql(table_name, engine, if_exists="replace", index=False)

print(f"Data successfully loaded into table '{table_name}' in database '{database}'.")
```

```
Data successfully loaded into table 'customer' in database 'Customer_behaviour'.
```

[ ]:

Figure: Relevant screenshot with explanation of transformation and output.

SQL: Revenue by Gender

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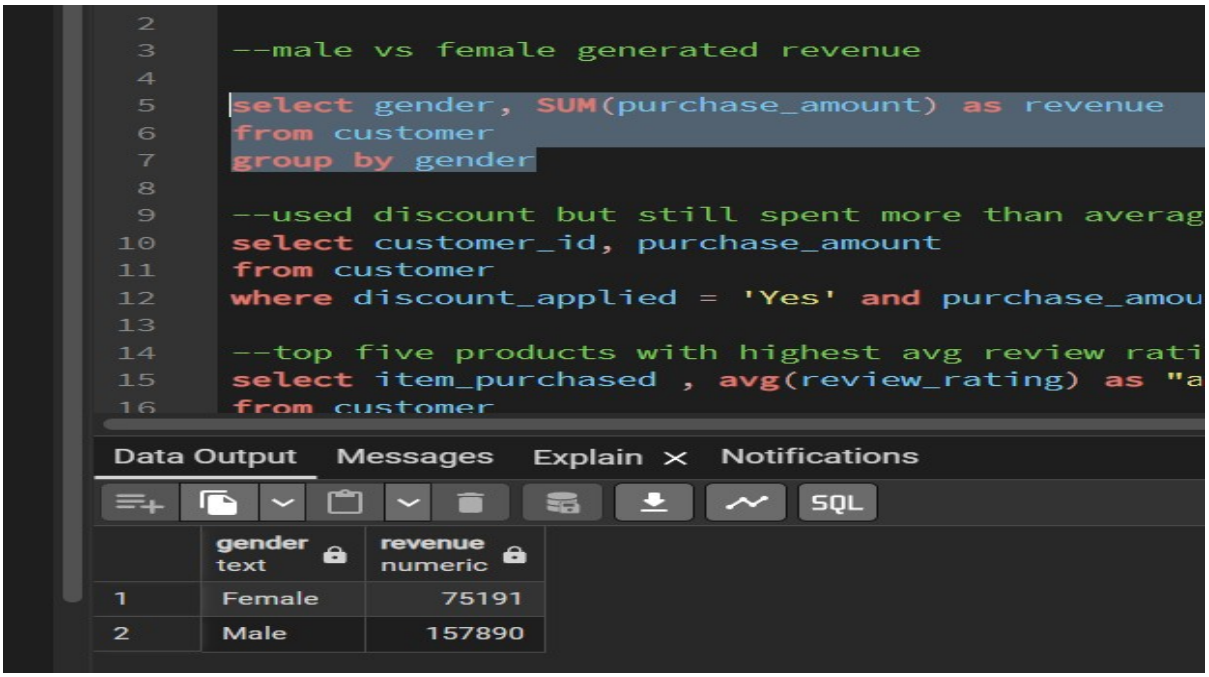

Figure: Relevant screenshot with explanation of transformation and output.

SQL: Discount High Spenders

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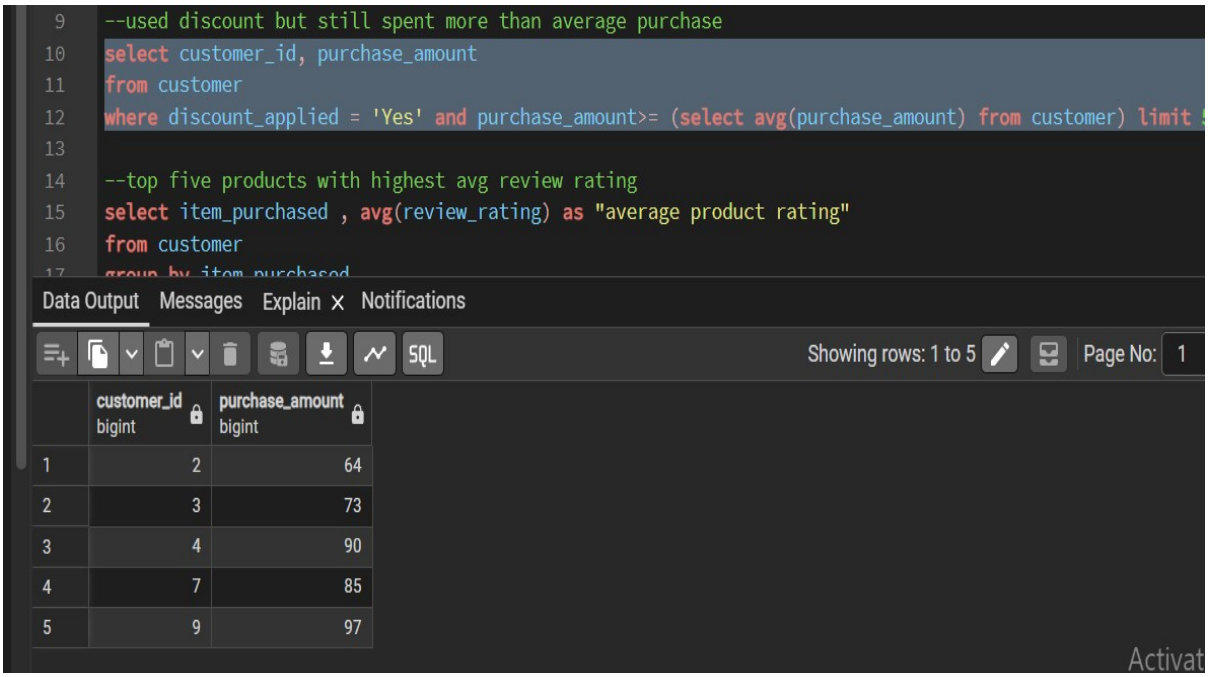

Figure: Relevant screenshot with explanation of transformation and output.

SQL: Revenue by Age Group

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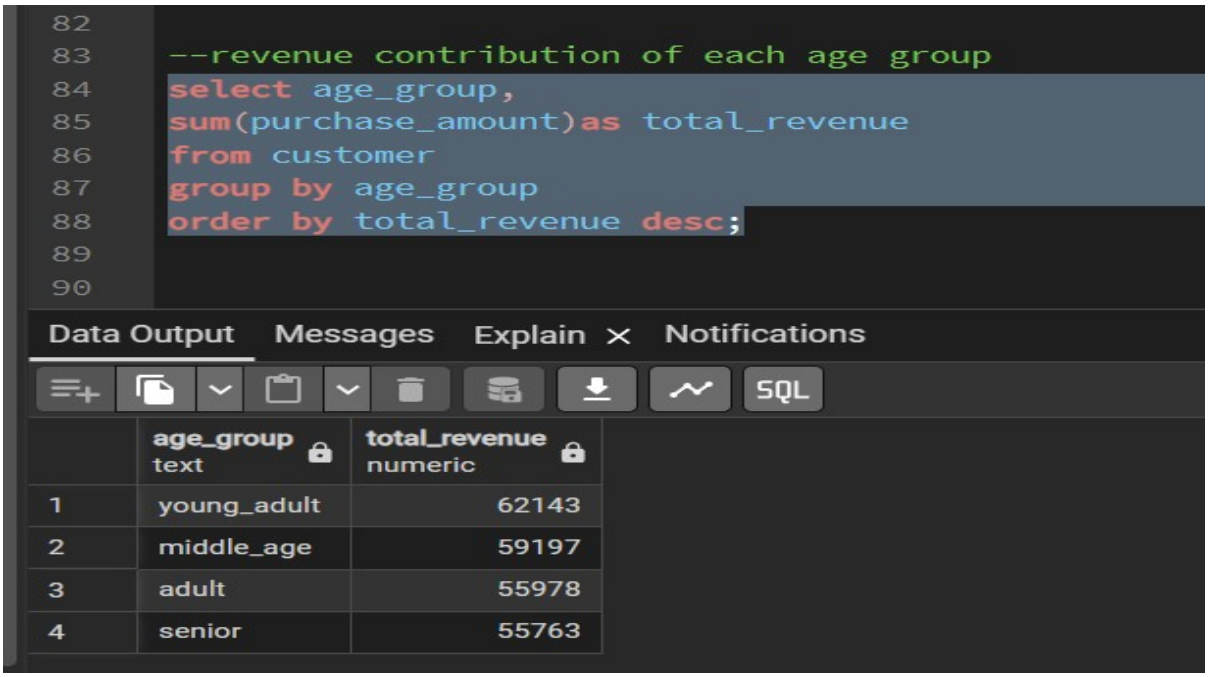

Figure: Relevant screenshot with explanation of transformation and output.

SQL: Subscribers Analysis

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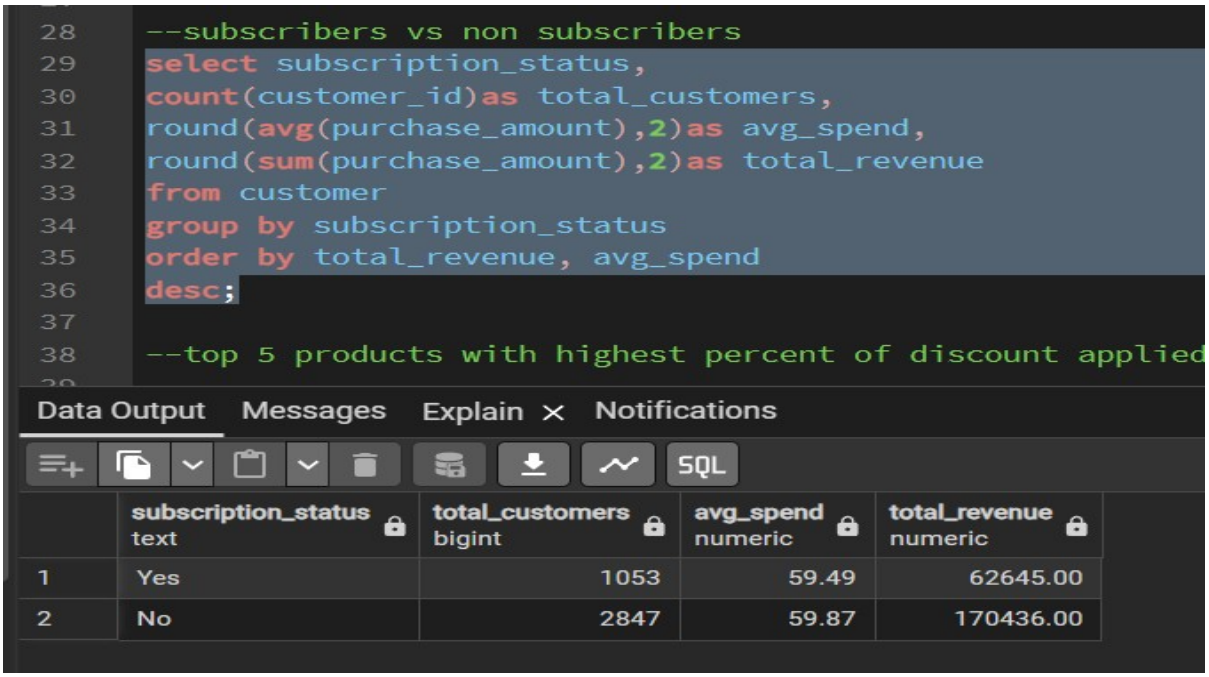

```
28  --subscribers vs non subscribers
29  select subscription_status,
30  count(customer_id)as total_customers,
31  round(avg(purchase_amount),2)as avg_spend,
32  round(sum(purchase_amount),2)as total_revenue
33  from customer
34  group by subscription_status
35  order by total_revenue, avg_spend
36  desc;
37
38  --top 5 products with highest percent of discount applied
```

Data Output | Messages | Explain | Notifications

SQL

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

Figure: Relevant screenshot with explanation of transformation and output.

SQL: Discount Rate

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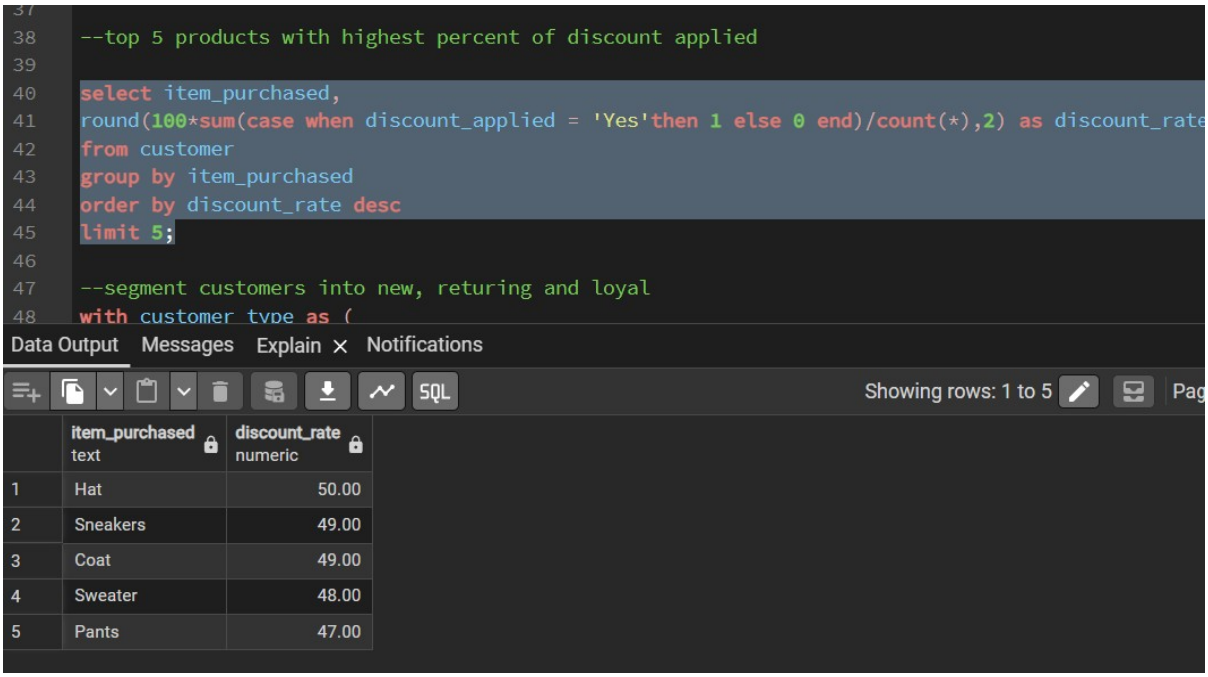

Figure: Relevant screenshot with explanation of transformation and output.

SQL: Product Ratings

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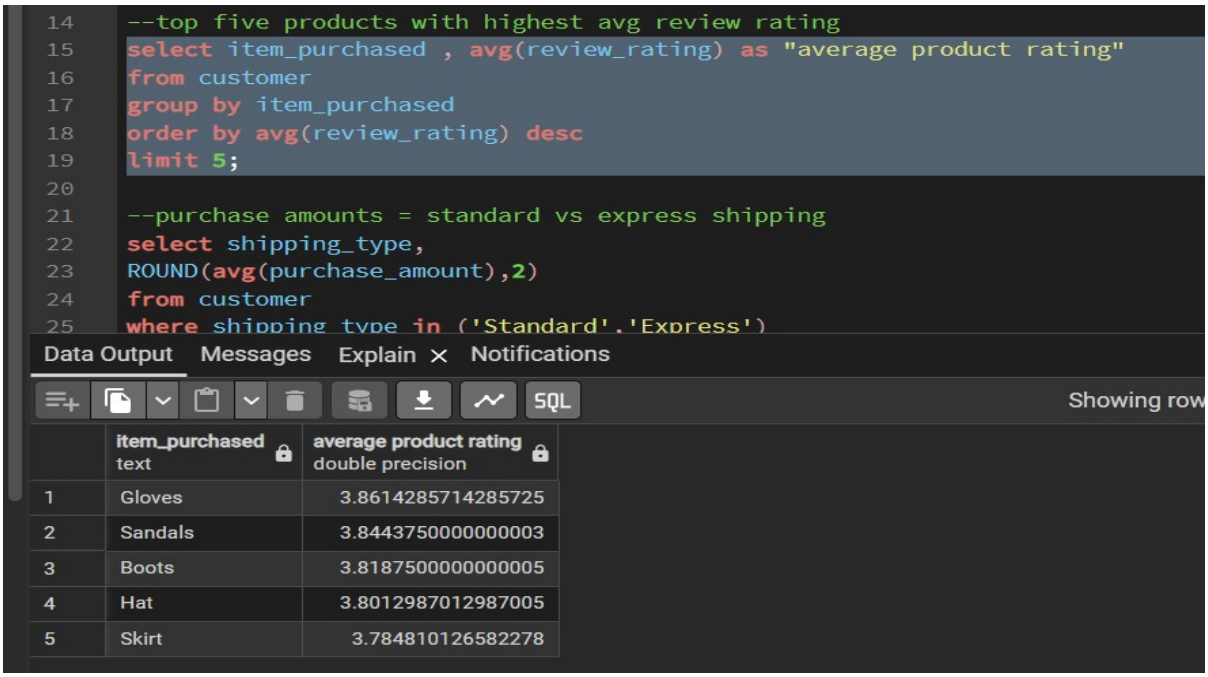

Figure: Relevant screenshot with explanation of transformation and output.

SQL: Customer Segmentation

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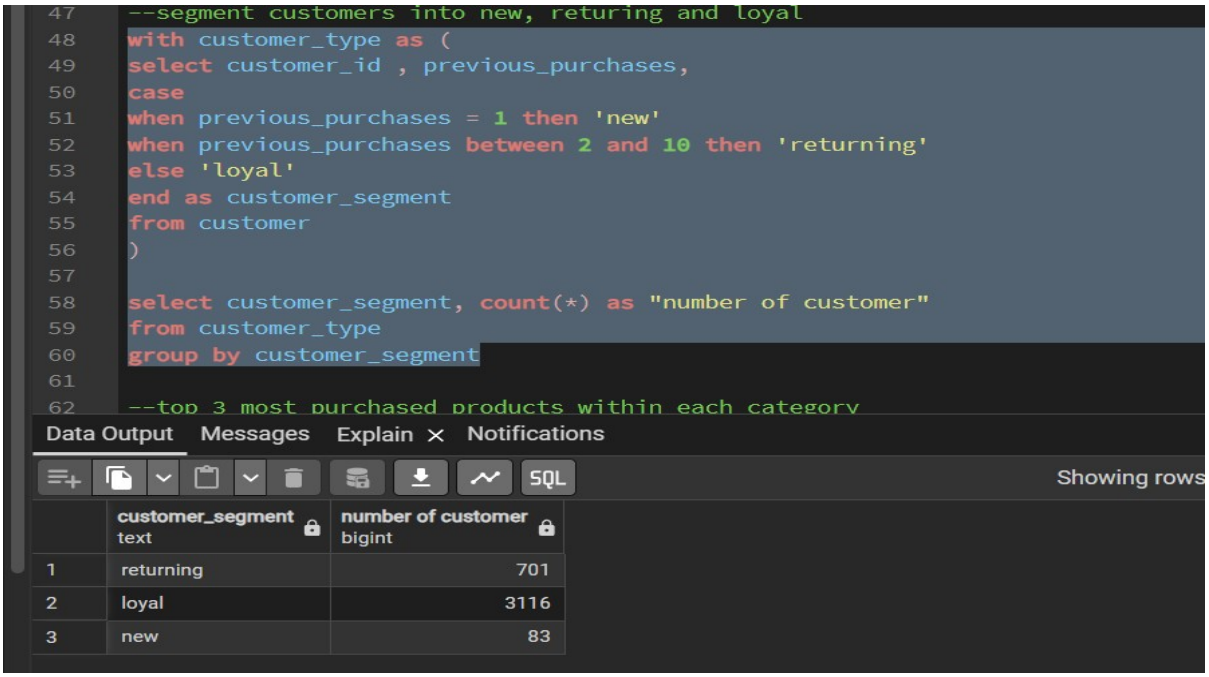

Figure: Relevant screenshot with explanation of transformation and output.

SQL: Shipping Analysis

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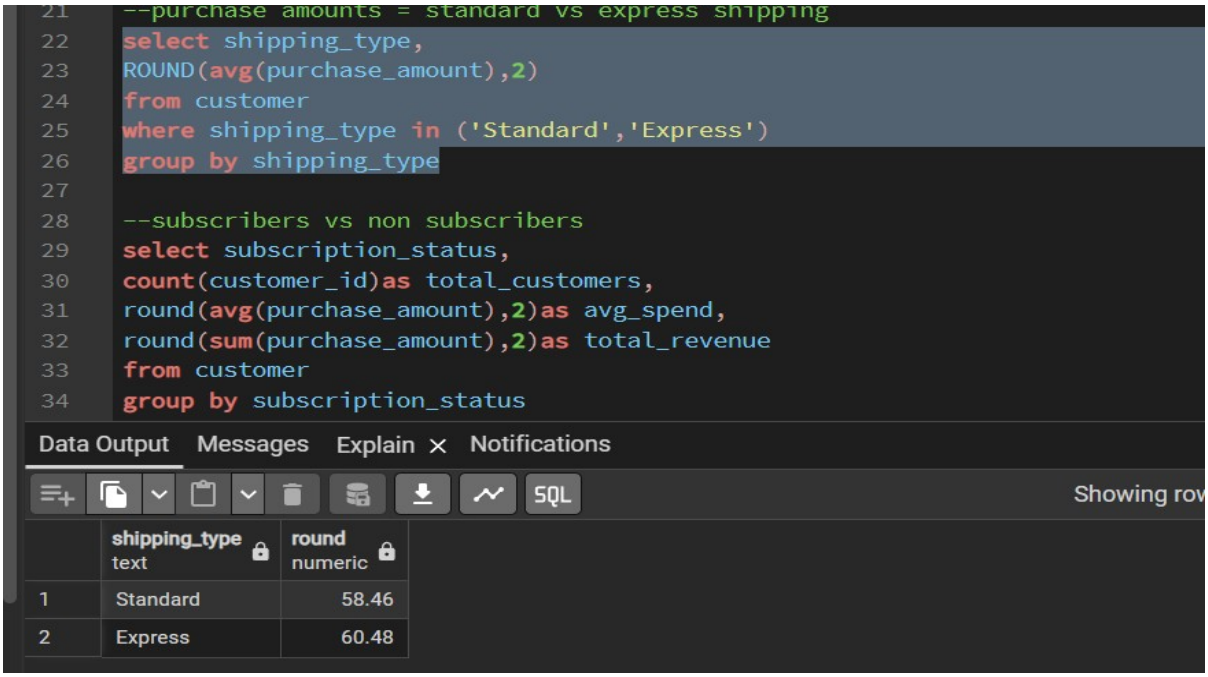

Figure: Relevant screenshot with explanation of transformation and output.

SQL: Top Products

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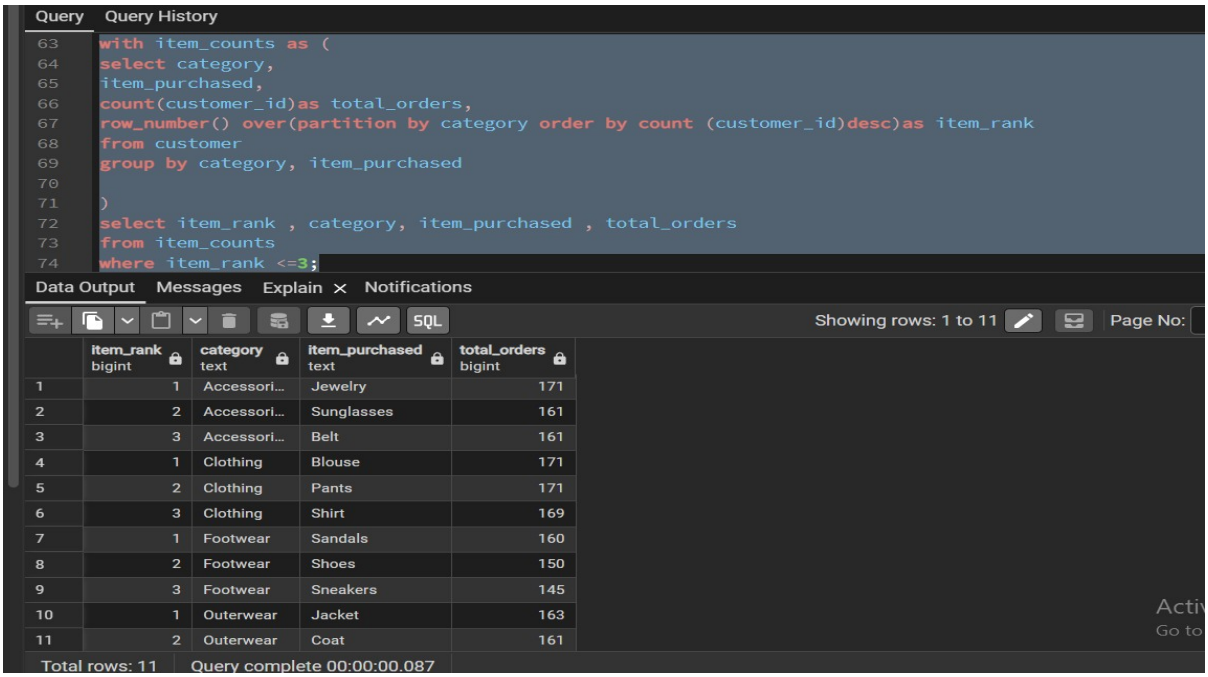

Figure: Relevant screenshot with explanation of transformation and output.

Power BI Dashboard Development & Insights

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Figure: Relevant screenshot with explanation of transformation and output.

Insights & Findings

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Business Recommendations

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Conclusion

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Future Scope

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